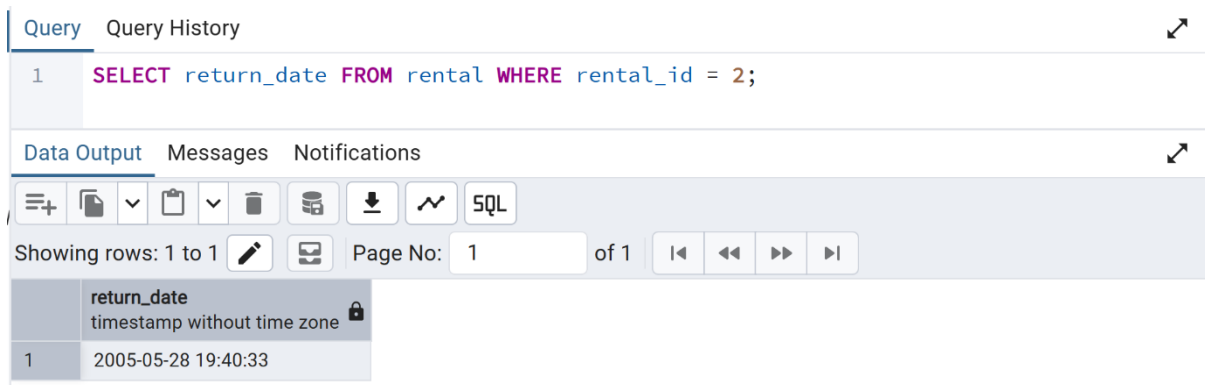


SECTION 5: TRANSACTIONS & ACID PROPERTIES

Lab 1: The Rental Refund Crisis.

Exercise 1: Testing the "All or Nothing"

- Step 1: Check data before execution.



Query Query History

```
1 SELECT return_date FROM rental WHERE rental_id = 2;
```

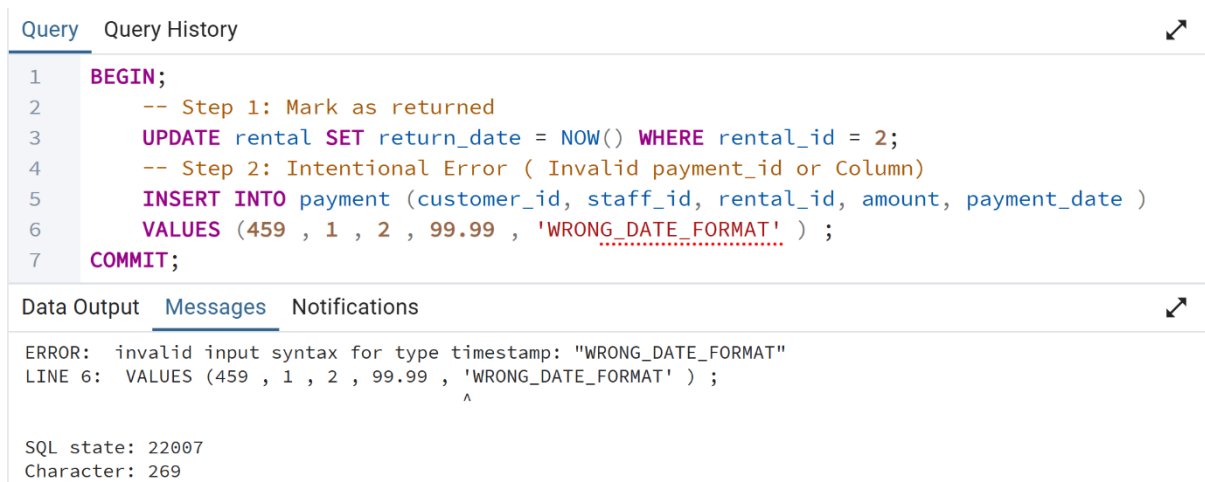
Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	return_date timestamp without time zone
1	2005-05-28 19:40:33

The returned result shows that the current disc return date is 2005-05-28 19:40:33.

- Step 2: Execute a transaction block containing an intentional error.



Query Query History

```
1 BEGIN;
2 -- Step 1: Mark as returned
3 UPDATE rental SET return_date = NOW() WHERE rental_id = 2;
4 -- Step 2: Intentional Error ( Invalid payment_id or Column)
5 INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date )
6 VALUES (459 , 1 , 2 , 99.99 , 'WRONG_DATE_FORMAT' ) ;
7 COMMIT;
```

Data Output Messages Notifications

ERROR: invalid input syntax for type timestamp: "WRONG_DATE_FORMAT"
LINE 6: VALUES (459 , 1 , 2 , 99.99 , 'WRONG_DATE_FORMAT') ;
^

SQL state: 22007
Character: 269

The screen shows error message: ERROR: invalid input syntax for type timestamp: "WRONG_DATE_FORMAT"

- Step 3: Check the results to see the effect of atomicity.

Query Query History



```
1 SELECT staff_id, first_name, last_name, store_id, username
2 FROM staff
3 WHERE username = 'vanh';
```

Data Output Messages Notifications



Showing rows: 1 to 1 Page No: 1 of 1

	staff_id [PK] integer	first_name character varying (45)	last_name character varying (45)	store_id smallint	username character varying (16)
1	6	Van	Anh	1	vanh